GAMURA - Google Play Store Launch Kit

Complete setup guide + AI task brief for Google AI Studio
Project: https://gamura.vercel.app/ | Founder: Selvaranjan G

0. INSTRUCTIONS FOR THE AI (READ FIRST)

You are a senior full-stack engineer and PWA / Android publishing expert. The project is **Gamura**, a Next.js web app deployed on Vercel with Firebase (Auth/Firestore) and built in Google AI Studio, pushed to GitHub. Goal: make Gamura a production-ready installable PWA that can be wrapped as an Android app (Trusted Web Activity) and published on Google Play, then monetized.

Do ALL of the following in the existing project, without breaking current features:

- Add/complete **public/manifest.json** (template in Section 3) with name Gamura, standalone display, theme color #10b981, background #0f172a, icons 192 and 512 (maskable too).
- Add a **service worker** (public/sw.js, Section 4) with offline fallback page and auto-update (skipWaiting, clients.claim, versioned cache). Register it in the app layout.
- Add **public/offline.html**, a simple branded offline page.
- Add **public/.well-known/assetlinks.json** using the template in Section 5. Leave the SHA-256 value as a clearly marked placeholder.
- Make sure Vercel serves assetlinks.json with Content-Type application/json (add headers in next.config.js / vercel.json if needed).
- Create pages **/privacy** and **/terms** (Section 6) and link them in the app footer/settings.
- Add meta tags: viewport, theme-color, apple-touch-icon, manifest link.
- Add an **Account Deletion** option/page (Google Play requires apps with login to offer account deletion and a web link for it). Delete Firebase Auth user + Firestore data.
- Add a **CHANGELOG.md** and a **PLAY_STORE_CHECKLIST.md** in the repo root.
- Run a Lighthouse-style self review and list anything still failing PWA install criteria.
- At the end, output a short list of files changed and the manual steps I still must do myself (Section 7 onward).

Rules: keep the existing Gamura emerald theme, do not remove features, do not hardcode secrets, keep all text professional.

1. Overview: The Plan

Wrap the Vercel website as an Android app using **PWABuilder** (Trusted Web Activity). No rewrite needed. Web updates go live instantly; the Play Store only needs a new upload when the Android wrapper itself changes.

Phase	What	Time
1	Make Gamura a PWA (Sections 3-6)	1-2 days
2	Generate AAB at pwabuilder.com	1 hour
3	Play Console account + store listing	1-2 days
4	Closed test: 12 testers, 14 days	14+ days

Phase	What	Time
5	Apply production + review	2-7 days

2. Important Play Store Rules

- Developer account: **$25 one-time**, ID verification required.
- Personal accounts created after 13 Nov 2023 must run a **closed test with at least 12 testers opted in for 14 continuous days** before applying for production. Organization accounts (with D-U-N-S number) are exempt. Testers must actually use the app.
- A privacy policy URL is mandatory (Gamura uses Firebase login).
- Apps with account creation must offer in-app and web account deletion.
- Do not submit a bare website with no value; Gamura has login, tools and games, so highlight them.
- Digital goods (coins, premium, subscriptions) sold in the Play app must use Google Play Billing.
- Do not target the app at children; keep audience 13+ or 18+.
- Rules change over time. Always re-check Play Console Help before submitting.

3. manifest.json (template)

```
{
  "name": "Gamura",
  "short_name": "Gamura",
  "description": "Gamura - games, tools and your digital identity in one place.",
  "start_url": "/?source=pwa",
  "scope": "/",
  "display": "standalone",
  "orientation": "portrait",
  "background_color": "#0f172a",
  "theme_color": "#10b981",
  "categories": ["games", "entertainment", "productivity"],
  "icons": [
    { "src": "/icons/icon-192.png", "sizes": "192x192", "type": "image/png" },
    { "src": "/icons/icon-512.png", "sizes": "512x512", "type": "image/png" },
    { "src": "/icons/maskable-512.png", "sizes": "512x512",
      "type": "image/png", "purpose": "maskable" }
  ]
}
```

4. Service worker (public/sw.js)

```
const CACHE = "gamura-v1";   // BUMP this on every release
const OFFLINE = "/offline.html";

self.addEventListener("install", (e) => {
  e.waitUntil(caches.open(CACHE).then((c) => c.addAll([OFFLINE, "/manifest.json"])));
  self.skipWaiting();
});

self.addEventListener("activate", (e) => {
  e.waitUntil(
    caches.keys().then((keys) =>
      Promise.all(keys.filter((k) => k !== CACHE).map((k) => caches.delete(k)))
    )
  );
  self.clients.claim();
});

self.addEventListener("fetch", (e) => {
  if (e.request.method !== "GET") return;
  if (e.request.mode === "navigate") {
    e.respondWith(fetch(e.request).catch(() => caches.match(OFFLINE)));
    return;
  }
  e.respondWith(
    fetch(e.request)
      .then((res) => {
        const copy = res.clone();
        caches.open(CACHE).then((c) => c.put(e.request, copy));
        return res;
      })
      .catch(() => caches.match(e.request))
  );
});
```

Register in the app layout (client side):

```
if ("serviceWorker" in navigator) {
  window.addEventListener("load", () => navigator.serviceWorker.register("/sw.js"));
}
```

5. assetlinks.json (domain verification)

Path: **public/.well-known/assetlinks.json**. Replace the package name and fingerprint. Get the SHA-256 from Play Console: Setup > App signing (use the App signing key certificate fingerprint). Add the upload key fingerprint too if testing locally.

```
[{
  "relation": ["delegate_permission/common.handle_all_urls"],
  "target": {
    "namespace": "android_app",
    "package_name": "com.gamura.app",
    "sha256_cert_fingerprints": ["PASTE:SHA256:FINGERPRINT:HERE"]
  }
}]
```

Optional vercel.json headers:

```
{ "headers": [{ "source": "/.well-known/assetlinks.json",
  "headers": [{ "key": "Content-Type", "value": "application/json" }] }] }
```

6. Privacy Policy and Terms (content outline)

Create /privacy and /terms pages in Gamura's style. Privacy policy must cover:

- Who we are: Gamura, founder Selvaranjan G, contact email (add your real email).
- Data collected: name, email, profile photo (Firebase Authentication / Google sign-in), content users create, basic analytics and device info.
- Why: account, features, security, improving the app.
- Third parties: Google Firebase, Vercel, Google Play services, and ad providers if added later (AdMob).
- Data retention and security.
- User rights: access, correction, deletion. Include the **account deletion** steps and URL (for example gamura.vercel.app/delete-account).
- Children: the app is not directed to children under 13.
- Changes to the policy and last updated date.

7. Manual steps YOU must do

A. Build the Android package

- Open **pwabuilder.com**, enter https://gamura.vercel.app/ and run the check.
- Choose Android > Google Play. Package ID e.g. com.gamura.app. Version code 1, version name 1.0.
- Let PWABuilder generate a signing key, then download the .aab and the signing key + password file. **Back them up safely. Never lose them.**

B. Play Console

- Create a developer account ($25) and verify your identity.
- Create app > Gamura > App or game: choose accordingly > Free.
- Upload the .aab to **Closed testing** (not production). Turn on Play App Signing.
- Copy the SHA-256 from App signing into assetlinks.json, push to GitHub, let Vercel deploy.
- Add 12+ testers (friends, classmates, family) by email list or Google Group; share the opt-in link. They must install and keep it for 14 days and open it sometimes.
- Complete: store listing, privacy policy URL, Data Safety form, content rating, target audience, ads declaration, app access (provide a test login if needed), account deletion URL.
- After 14 days, click **Apply for production**. Review takes about 2-7 days.

C. Store listing assets

Asset	Spec
App icon	512 x 512 PNG, max 1 MB
Feature graphic	1024 x 500 PNG/JPG
Phone screenshots	At least 4, 16:9 or 9:16, min 320px
Short description	Max 80 characters
Full description	Max 4000 characters

Suggested short description: Gamura - games, tools and your digital identity in one app.

8. Updating the app after launch

Normal update (no Play Store step)

- Edit in Google AI Studio > push to GitHub > Vercel auto-deploys > users see it next open.
- Bump the service worker CACHE name every release so users never get stale files.

When a new Play upload IS needed

- Changing icon/name/splash, adding permissions or push notifications.
- Switching to Capacitor for AdMob or native billing.
- Google's yearly target API level requirement (rebuild about once a year).
- Steps: regenerate at pwabuilder.com > **increase version code** (2, 3, ...) and version name > sign with the SAME key > Play Console > Production > Create new release > upload .aab > release notes > submit.

Store listing text, screenshots and policy links can be edited anytime in Play Console (short review). If you add new data collection, update the **Data Safety form** too.

9. Monetization Plan

Method	How	Notes
Ads	AdMob via a Capacitor wrapper	AdSense inside a TWA/WebView can violate policy. Use AdMob natively.
In-app purchases	Google Play Billing (Digital Goods API in TWA)	Required for digital items sold in the Play version.
Subscription	Gamura Premium: ad-free, themes, extra features	Best recurring income.
Gamura ID Pro	Custom URLs, portfolio themes	Good for students and creators.
Sponsors / affiliates	Placements in Gamura Galaxy	Needs traffic first.

Google charges a service fee on Play Billing sales (check current rates in Play Console). Payouts need a merchant profile with bank and tax details.

Growth strategy

- Launch free with light ads or no ads.
- Reach about 1,000 active users.
- Add Premium subscription and a referral reward system.
- Post updates on LinkedIn/GitHub, share Play link in college groups, collect reviews (ask happy users politely).
- Track retention in Firebase Analytics and improve the top 3 most used features.

10. Final Checklist

- [ ] manifest.json, icons (192, 512, maskable)
- [ ] sw.js + offline.html registered
- [ ] assetlinks.json live at /.well-known/ with correct SHA-256
- [ ] /privacy, /terms, account deletion page live

- [ ] .aab built and backed up with signing key
- [ ] Play Console account verified
- [ ] Closed test running with 12+ opted-in testers for 14 days
- [ ] Store listing, Data Safety, content rating, audience done
- [ ] Production access applied and approved
- [ ] CHANGELOG.md started

Note: Play Store policies and fees change. Verify current requirements in Google Play Console Help before each submission.